

Diabetic Retinopathy Grading on IDRiD: A Comparative Study of Architectures, Preprocessing, and Ensemble Strategies

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Abstract

This study compares ResNet-18, EfficientNet-B0, EfficientNet-B2, and ViT-B/16 for five-class diabetic retinopathy grading on the IDRiD dataset. The study investigate transfer learning, class imbalance handling, fundus-specific preprocessing, and ensemble strategies. EfficientNet-B2 achieved the strongest single-model performance with a test Macro F1 of 0.5273. The best Vision Transformer reached 0.4679, while the final ensemble improved the test Macro F1 to 0.5823. The results show that targeted preprocessing can benefit individual models, whereas combining models with complementary error patterns provides the most balanced performance. The observed validation-test differences also demonstrate the risk of overfitting on small medical imaging datasets.

Keywords

Deep Learning — Diabetic Retinopathy — EfficientNet — Vision Transformer — Image Preprocessing — Ensemble Learning

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1. Introduction

Diabetic retinopathy (DR) is a progressive complication of diabetes and an important cause of preventable visual impairment. Its severity is commonly represented by ordered stages ranging from no apparent retinopathy to proliferative diabetic retinopathy [1]. Retinal fundus photography enables non-invasive examination of disease-related changes, but manual assessment requires clinical expertise and becomes difficult to scale in large screening programs. Deep learning has there-

fore gained considerable attention as a tool for supporting automated analysis of retinal images. Previous work has demonstrated that neural networks can achieve strong performance in detecting referable diabetic retinopathy from fundus photographs [2].

Publicly available and consistently annotated datasets are necessary for developing and comparing such methods. The Indian Diabetic Retinopathy Image Dataset (IDRiD) provides retinal fundus images with annotations for lesion segmentation, disease grading, and diabetic macular edema [3]. Its disease-grading task defines five ordinal DR classes and was also used in the IDRiD Segmentation and Grading Challenge [4]. However, the grading subset is relatively small and strongly imbalanced. In particular, mild diabetic retinopathy is represented by considerably fewer images than the most frequent classes. Furthermore, visual differences between neighboring severity grades may be subtle. These properties make model selection and reliable evaluation difficult: a model can achieve reasonable overall accuracy while still performing poorly on rare classes.

This report investigates several deep-learning strategies for five-class diabetic retinopathy grading on IDRiD. The experiments compare the convolutional architectures ResNet-18, EfficientNet-B0, and EfficientNet-B2 with a Vision Transformer, ViT-B/16. In addition to the model architecture, the study examines transfer learning with frozen and fully fine-tuned backbones, weighted loss functions, sampling and augmentation strategies, fundus-specific preprocessing, hyperparameter optimization, and model ensembles. Macro F1 is used as the primary evaluation metric because it assigns equal importance to every class, while quadratic weighted kappa is reported as a complementary measure that considers the ordinal structure of the disease grades.

Accordingly, this report addresses the following research question: *How do CNN-based architectures, Vision Transformers, preprocessing strategies, and ensemble methods compare for diabetic retinopathy grading on the IDRiD dataset?*

The main contribution of this work is a structured empirical comparison of these approaches under a shared evaluation protocol. The study further analyzes class-specific errors and examines whether models with complementary behavior can be combined to improve the balance of predictions across all five disease grades.

2. Background and Related Work

Diabetic retinopathy grading is commonly formulated as an ordinal multi-class classification problem. In the IDRiD disease-grading task, each retinal fundus image is assigned to one of five grades, ranging from no apparent diabetic retinopathy to proliferative diabetic retinopathy [3, 4]. This ordinal structure is important: confusing two neighboring grades is usually

less severe than confusing a healthy retina with a proliferative case. At the same time, the visual differences between adjacent stages can be subtle, especially when early lesions occupy only small regions of the image. These characteristics explain why diabetic retinopathy grading is more difficult than a binary screening task, even though deep learning has already shown strong performance for detecting referable diabetic retinopathy in retinal photographs [2].

Previous work on diabetic retinopathy grading has often relied on convolutional neural networks, sometimes combining several CNNs to increase robustness. For example, Kori et al. proposed an ensemble of CNNs for automatic grading of diabetic retinopathy and diabetic macular edema, showing that combining models can be useful when individual networks make different errors [5]. This idea is relevant for the present project because the IDRiD grading set is small and imbalanced, making individual models sensitive to validation-split noise. More recent work has also explored transformer-based approaches for retinal disease analysis, motivated by the ability of self-attention to model long-range image relationships [?].

The CNN architectures used in this study represent two different design principles. ResNet introduced residual connections, where layers learn residual mappings through shortcut connections rather than only stacked transformations [6]. This design makes it easier to train deeper networks and is therefore a common baseline architecture for image classification. EfficientNet follows a different strategy: instead of scaling depth, width, or image resolution independently, it uses compound scaling to increase these dimensions in a balanced way [7]. EfficientNet-B0 is the smallest baseline variant, while EfficientNet-B2 is a larger variant with increased capacity and input resolution. In this project, B0 is therefore used as a compact comparison model, whereas B2 serves as the stronger CNN candidate.

In contrast, the Vision Transformer represents an image as a sequence of patches and processes these patch embeddings with a transformer encoder [8]. Its attention mechanism is based on the transformer architecture introduced by Vaswani et al. [9]. While CNNs include strong local inductive biases through convolutional filters, ViTs rely more heavily on data and pretraining to learn useful spatial relationships. This difference is especially relevant for medical imaging datasets such as IDRiD, where the number of labeled images is limited.

For this reason, transfer learning is an important part of the experimental design. Instead of training all models from random initialization, ImageNet-pretrained weights are used and then adapted to the retinal grading task. Prior work has shown that fine-tuning pretrained CNNs can be effective in medical image analysis, but the best strategy depends on the dataset size, similarity to the pretraining domain, and amount of available supervision [10]. More generally, pretrained neural features are transferable to new tasks to different degrees, with earlier

layers often capturing more generic visual patterns and later layers becoming more task-specific [11]. Therefore, this report compares frozen-backbone settings with full fine-tuning and treats the degree of adaptation as an experimental factor rather than a fixed assumption.

3. Data and Evaluation Protocol

The experiments use the disease-grading part of the Indian Diabetic Retinopathy Image Dataset (IDRiD). IDRiD provides retinal fundus photographs and annotations for several diabetic retinopathy tasks, including the five-class disease-grading task used in this report [3, 4]. The official grading data contains 413 training images and 103 test images.

Because the official test set must not be used for model selection, the official training set was split into an internal training and validation set. A stratified split with seed 42 and a validation fraction of 20% produced 330 training images and 83 validation images. The official test set was only evaluated after selecting a model or configuration based on validation performance.

Table 1. Dataset splits used in the experiments.

Split	Images
Training	330
Validation	83
Official test	103

The three splits were used for different purposes: model fitting, checkpoint and configuration selection, and final evaluation. This separation was kept fixed across the main experiments.

The class distribution is strongly imbalanced, which is one of the central challenges of the project. The five labels correspond to increasing disease severity from grade 0 to grade 4. In the official training set, grade 1 is represented by only 20 images. After the internal split, only 16 grade-1 images remain in the training set and only four in the validation set. The official test set contains only five grade-1 samples. Consequently, a small number of changed predictions can noticeably change class-wise F1 and Macro F1. This issue is typical for imbalanced learning problems, where overall accuracy can be dominated by frequent classes [12].

Table 2. Class distribution after the internal stratified split.

Split	0	1	2	3	4
Training	107	16	109	59	39
Validation	27	4	27	15	10
Official test	34	5	32	19	13

Accuracy, Macro F1, quadratic weighted kappa, class-wise F1 scores, and confusion matrices were used to evaluate the models. Accuracy measures the proportion of correctly classified images. While intuitive, it is not sufficient as the main

metric for this dataset because the majority classes have a much larger influence on the score than rare classes. A model can therefore obtain reasonable accuracy while still failing to identify grade 1 cases.

For this reason, Macro F1 was used as the primary selection and comparison metric. For each class, precision and recall are computed from the corresponding one-vs-rest classification problem. The F1 score is the harmonic mean of precision and recall:

$$F1_c = 2 \cdot \frac{\text{Precision}_c \cdot \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}. \quad (1)$$

Macro F1 is then calculated as the unweighted average across all C classes:

$$\text{Macro F1} = \frac{1}{C} \sum_{c=1}^C F1_c. \quad (2)$$

In this study, $C = 5$. Because every class contributes equally, Macro F1 is more sensitive to poor performance on rare classes than accuracy or weighted F1. This makes it suitable for comparing models under the strong imbalance shown in Table 2.

Quadratic weighted kappa was used as a secondary metric because diabetic retinopathy grades are ordinal. It measures agreement between predicted and true labels while accounting for agreement expected by chance. With quadratic weights, larger disagreements are penalized more strongly than small disagreements [13]. Thus, predicting grade 3 instead of grade 4 is treated as less severe than predicting grade 0 instead of grade 4. This complements Macro F1: Macro F1 emphasizes balanced class-wise recognition, whereas quadratic weighted kappa reflects the ordered nature of the severity scale.

Finally, confusion matrices and per-class precision, recall, and F1 scores were used for error analysis. These metrics are important because two models can have similar Macro F1 values while making different types of errors. In particular, the experiments later show that some models improve higher-grade recognition or kappa while others provide better balance across all five classes. The validation set was used for checkpoint selection, early stopping, hyperparameter choices, and ensemble selection. Test metrics were reported only after these decisions had been made.

4. Methodology

4.1 Shared Experimental Pipeline

All experiments followed the same general pipeline. Each fundus image was loaded from the IDRiD grading dataset,

optionally transformed by a preprocessing or augmentation step, resized to the model-specific input resolution, and passed through a pretrained backbone. The original classification layer was replaced by a task-specific head with five output neurons, one for each diabetic retinopathy grade. The final prediction was obtained by selecting the class with the highest model output score.

ImageNet-pretrained weights were used for all main architectures. This was motivated by the small size of the IDRiD grading set and by previous work showing that fine-tuning pretrained models can be effective in medical image analysis [10, 11]. Depending on the experiment, the pretrained backbone was either frozen or fully fine-tuned. A frozen backbone only updates the classification head, while full fine-tuning updates both the head and the pretrained feature extractor.

The validation set was used for checkpoint selection, early stopping, hyperparameter choices, and ensemble selection. Unless an experiment intentionally tested a different setting, the best checkpoint was selected by validation Macro F1. This choice aligns the training objective with the primary evaluation metric introduced in Section 3. Training was stopped early when the validation score did not improve for a fixed patience window. The official test set was only evaluated after the model configuration had been selected.

The early development phase used Optuna for exploratory hyperparameter search [14]. These searches helped identify plausible ranges for learning rate, weight decay, dropout, optimizer choice, batch size, and loss function. However, because the validation set is small, later experiment matrices used fixed, interpretable settings whenever possible. This made comparisons between model variants more reproducible and reduced the risk of comparing heavily tuned runs against lightly tuned ones. AdamW was used in many experiments as a weight-decay-regularized optimizer [15]. Training histories and final metrics were retained for later comparison.

4.2 CNN Models

Three convolutional architectures were evaluated: ResNet-18, EfficientNet-B0, and EfficientNet-B2. ResNet-18 served as the first CNN baseline because it is compact, widely used, and based on residual connections [6]. The ResNet experiments compared a frozen backbone, weighted losses, oversampling, augmentation, Graham-style preprocessing, and full fine-tuning. This produced a controlled baseline matrix for understanding which factors had a visible effect before moving to larger CNNs.

EfficientNet-B0 was used as a compact EfficientNet baseline. EfficientNet-B2 was then tested as a higher-capacity model. EfficientNet models scale network width, depth, and input resolution in a more balanced way than simply increasing one dimension independently [7]. In this project, EfficientNet-B2 became the main CNN candidate because it showed the

strongest single-model performance in the later experiments. The CNN models were generally trained with an input resolution of 384×384 pixels.

For the CNNs, the classification head was adapted to the five-class task. In most later experiments, a custom head with a hidden layer, nonlinear activation, normalization, dropout, and final linear output layer was used. Dropout was included to reduce overfitting on the small dataset. The exact hyperparameters differed between experiment matrices, but all final results were exported together with their configuration values to make the runs reproducible.

4.3 Vision Transformer

The transformer-based comparison model was ViT-B/16. Instead of applying convolutional filters over local neighborhoods, ViT-B/16 divides the image into fixed-size patches and processes their embeddings with a transformer encoder [8]. The self-attention mechanism is based on the transformer architecture introduced by Vaswani et al. [9]. This makes the model conceptually different from the CNNs, since global interactions between image regions can be modeled through attention.

The ViT experiments used ImageNet-pretrained torchvision weights and an input size of 224×224 pixels. The ViT pipeline used weighted cross entropy and AdamW, and tested green-channel and masking-based preprocessing variants. The strongest reportable ViT configuration used the `clean_green_masked` preprocessing variant. An Optuna-based ViT tuning run was also performed, but its high validation score did not generalize well to the test set. In the report, this run is therefore treated as evidence of validation instability rather than as the final ViT model.

4.4 Preprocessing and Augmentation

Several fundus-specific preprocessing methods were evaluated because retinal images can contain dark borders, uneven illumination, and low local contrast. The simplest setting used no manual preprocessing beyond resizing and standard normalization. Additional preprocessing variants included cropping the retinal region, circular masking, autocontrast, Graham-style illumination correction, grayscale conversion with gamma correction and CLAHE, green-channel extraction, and green-channel CLAHE with background subtraction and a field-of-view mask.

CLAHE and related adaptive contrast methods are intended to enhance local contrast while limiting excessive amplification of noise [16]. Green-channel preprocessing was considered because retinal structures and vessels are often more visible in the green channel than in the red or blue channels. Related retinal preprocessing work also motivates illumination correction and contrast enhancement for fundus images [17, 18]. The exact Graham-style procedure used here should be under-

Table 3. Overview of the main architectures and their role in the study.

Architecture	Input size	Pretraining	Backbone modes	Main role
ResNet-18	384×384	ImageNet	Frozen / fine-tuned	CNN baseline
EfficientNet-B0	384×384	ImageNet	Frozen / fine-tuned	Compact CNN comparison
EfficientNet-B2	384×384	ImageNet	Mostly fine-tuned	Main CNN candidate
ViT-B/16	224×224	ImageNet	Fine-tuned	Transformer comparison

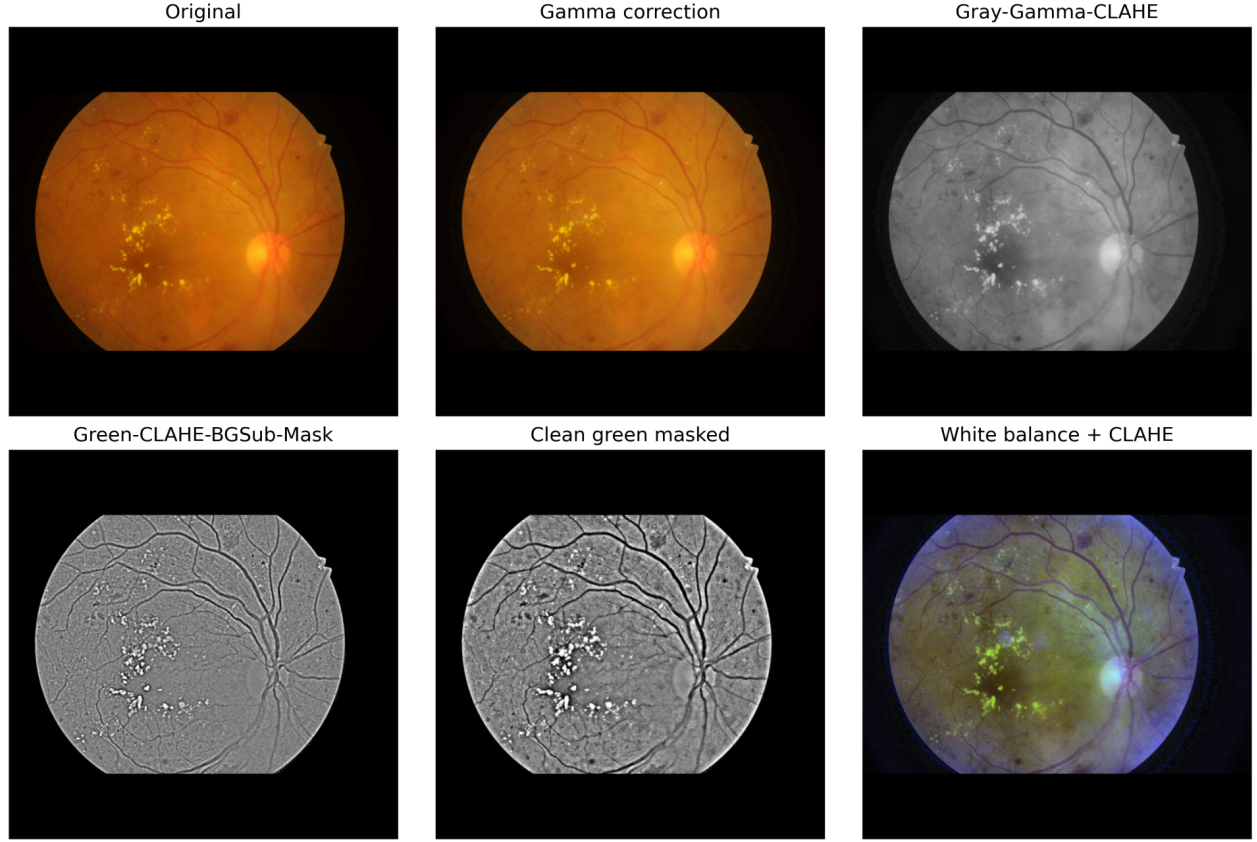


Figure 1. Comparison of selected preprocessing variants applied to the same fundus image.

stood as a project-specific implementation rather than a single canonical published method.

Data augmentation was tested as an additional way to improve robustness. The augmentation settings included mild random flips, small rotations, color jitter, random resized crops, and random erasing in selected runs. These transformations were intentionally kept moderate because aggressive augmentation can alter clinically relevant visual cues. Image augmentation is commonly used to improve generalization in deep learning, especially when training data are limited [19]. However, augmentation was treated as an experimental factor rather than assumed to be beneficial in every configuration.

4.5 Class-Imbalance Strategies

The class distribution described in Section 3 motivated several imbalance-handling strategies. The experiments compared standard cross entropy with weighted cross entropy, where

classes with fewer samples receive larger loss weights. Focal loss was also tested during the exploratory tuning phase. Focal loss reduces the relative contribution of easy examples and focuses learning more strongly on difficult samples [20]. In addition, some runs used weighted oversampling so that rare classes appeared more often during training.

These strategies were evaluated carefully because imbalance handling can change the trade-off between metrics. For example, a method may improve minority-class recall but reduce precision or increase confusion between neighboring grades. Therefore, Macro F1, quadratic weighted kappa, per-class F1, and confusion matrices were all inspected instead of relying on accuracy alone. Macro F1 was used as the main check-point and comparison metric because it gives each class equal weight.

4.6 Ensemble Methods

After training individual models, several ensemble strategies were evaluated. The motivation was that different architectures and preprocessing variants produced different error patterns. Even a model with weaker overall Macro F1 can still be useful if it correctly predicts samples that stronger models miss. This is the main reason why ensembles were explored after the single-model experiments.

The simplest ensemble method was majority voting, where each model predicts one class and the class with the most votes is selected. Soft voting averaged predicted class probabilities across models. Weighted soft voting assigned larger weights to stronger validation models. Optimized weighted soft voting searched for probability weights on the validation set. A class-adjusted soft vote was also tested to compensate for class-specific weaknesses. Finally, logit-level fusion methods were evaluated, including raw mean logits, weighted logits, z-score-normalized logits, and temperature-scaled logits. Temperature scaling is commonly used as a calibration method for neural network probabilities [21].

All ensemble combinations and weights were selected using the validation set. The official test set was used only for the final evaluation of selected ensemble candidates. Exploratory results that were sorted by test performance were not treated as final model-selection results. This distinction is important because validation-based weight optimization can overfit when the validation set is small. The ensemble experiments were motivated both by diabetic-retinopathy-specific CNN ensemble work [5] and by the general observation that diverse deep ensembles can improve robustness [22].

5. Results

This section summarizes the experimental results across CNN architectures, preprocessing variants, Vision Transformer runs, and ensemble methods. The main comparison metric is test Macro F1, while quadratic weighted kappa is reported as a complementary metric for the ordinal grading structure. Unless stated otherwise, all values refer to the official IDRiD test set.

5.1 CNN Architecture and Training Comparison

Table 4 reports the strongest representative result for each CNN architecture. The complete ResNet-18, EfficientNet-B0, and EfficientNet-B2 experiment matrices are provided in the appendix. Among the CNNs, EfficientNet-B2 achieved the strongest single-model result before additional preprocessing, reaching a test Macro F1 of 0.5157. ResNet-18 and EfficientNet-B0 performed clearly lower by Macro F1, although EfficientNet-B0 later contributed useful diversity in ensembles.

Table 4. Best representative CNN results on the official test set.

Model	Config.	Acc.	F1	Kappa
ResNet-18	R7	0.5243	0.4122	0.5663
EfficientNet-B0	B0-7	0.5340	0.3810	0.6146
EfficientNet-B2	B2-4	0.5825	0.5157	0.6007

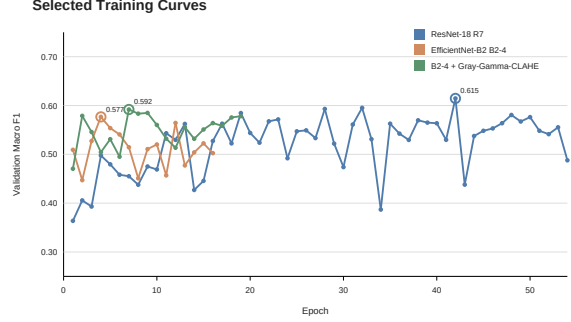


Figure 2. Validation Macro F1 over epochs for selected CNN runs.

Table 5. Selected EfficientNet-B2 preprocessing results.

Base	Preprocessing	F1	Kappa
B2-4	None	0.5157	0.6007
B2-4	Gray-Gamma-CLAHE	0.5273	0.6169
B2-6	Gray-Gamma-CLAHE	0.4984	0.6967
B2-4	Green-CLAHE-BGSub	0.5044	0.6681
B2-4	Gamma only	0.4452	0.6475

The ResNet experiments showed that full fine-tuning was more effective than keeping the backbone frozen. The best ResNet run, R7, used Crop+Graham preprocessing and full fine-tuning. EfficientNet-B0 showed a similar trend: B0-7, the fully fine-tuned configuration, achieved the best B0 Macro F1. EfficientNet-B2 was the strongest CNN family overall. In particular, B2-4, which used no manual fundus preprocessing, outperformed the Crop+Graham B2 variants. This was an important observation because it showed that additional preprocessing was not automatically beneficial.

Figure 2 shows selected validation curves for R7, B2-4, and B2-4 with Gray-Gamma-CLAHE. The curves illustrate that the best validation checkpoint can occur relatively early and that validation performance is unstable on the small validation split.

5.2 Preprocessing and Imbalance Handling

After EfficientNet-B2 became the strongest single-model family, additional preprocessing variants were tested. Table 5 lists the most relevant B2 preprocessing results. Gray-Gamma-CLAHE improved B2-4 from 0.5157 to 0.5273 test Macro F1 and became the strongest individual model in the study. However, the effect of preprocessing depended strongly on the underlying training configuration.

The strongest Macro F1 came from B2-4 with Gray-Gamma-

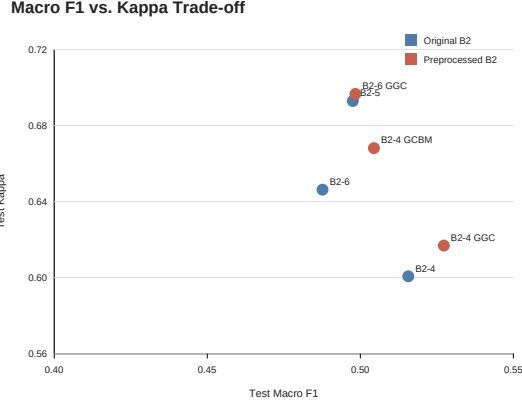


Figure 3. Trade-off between test Macro F1 and quadratic weighted kappa for selected B2 configurations.

Table 6. ViT-B/16 results on the official test set.

Configuration	Val F1	Acc.	F1	Kappa
Green-CLAHE-mask	0.5804	0.4951	0.4303	0.6692
Clean-green-mask	0.5804	0.5340	0.4679	0.6832
Optuna final	0.6817	0.3981	0.3220	0.5903

CLAHE, while B2-6 with Gray-Gamma-CLAHE achieved the strongest kappa among the selected preprocessing runs. This suggests that different preprocessing and loss configurations shifted the balance between class-wise recognition and ordinal agreement. For example, B2-6 used weighted cross entropy and produced a lower Macro F1 than B2-4 with Gray-Gamma-CLAHE, but achieved a considerably higher kappa. In contrast, gamma correction alone had a high validation Macro F1 but a weaker test Macro F1, indicating a validation-test gap.

Figure 3 visualizes this trade-off. B2-4 with Gray-Gamma-CLAHE achieved the best Macro F1, whereas B2-6 with Gray-Gamma-CLAHE achieved the strongest kappa among the selected preprocessing variants.

5.3 Vision Transformer Results

The Vision Transformer experiments used ViT-B/16 with ImageNet-pretrained weights and green-channel or masking-based preprocessing. Table 6 summarizes the most relevant ViT results. The strongest reportable ViT configuration was `clean_green_masked`, with a test Macro F1 of 0.4679 and a test kappa of 0.6832. This was below the best EfficientNet-B2 single model by Macro F1, but competitive by kappa.

The Optuna-tuned ViT run illustrates the difficulty of model selection on the small validation set. Although the best tuning score reached 0.6817 on validation, the final test Macro F1 was only 0.3220. This large validation-test gap suggests that the selected hyperparameters did not generalize reliably. Therefore, the `clean_green_masked` configuration is treated as the strongest reportable ViT result.

Table 7. Main ensemble results on the official test set.

Ensemble	Acc.	F1	Kappa
V1 majority vote	0.6602	0.5732	0.6917
V3 class-adjusted soft vote	0.6311	0.5823	0.7036
V3 weighted soft vote	0.6117	0.5443	0.7141
Optimized weighted soft vote	0.6311	0.5616	0.7546
Z-score mean logits	0.6214	0.5430	0.7070

Table 8. Class-wise F1 comparison on the official test set.

Model	0	1	2	3	4
ViT clean	0.6957	0.2857	0.4828	0.5116	0.3636
V3 ensemble	0.7500	0.4444	0.5862	0.5789	0.5517

5.4 Ensemble Results

The ensemble experiments produced the largest overall improvement. Table 7 compares the most relevant ensemble methods. The first five-model majority-vote ensemble already improved the test Macro F1 to 0.5732. The final V3 class-adjusted soft vote achieved the best validation-selected Macro F1 of 0.5823. The later optimized weighted soft vote from the logit-fusion notebook achieved the highest observed validation-selected kappa of 0.7546, but its Macro F1 was lower than the V3 class-adjusted ensemble.

The ensemble results show that model diversity was more important than simply selecting the single strongest architecture. Several models with lower individual Macro F1 still contributed useful complementary predictions. In particular, the final V3 class-adjusted soft vote combined EfficientNet-B2, EfficientNet-B0, ResNet-18, and preprocessing variants. Its per-class F1 profile was more balanced than that of the strongest individual models, especially for the rare classes.

Logit-level fusion did not improve over the strongest probability-based ensemble. The best pure logit method by Macro F1 was z-score-normalized mean logits with a test Macro F1 of 0.5430. Raw mean logits achieved a slightly higher kappa of 0.7149, but still did not exceed the optimized probability ensemble by kappa or the V3 class-adjusted ensemble by Macro F1.

Figure 4 summarizes the main performance milestones. The best single model was B2-4 with Gray-Gamma-CLAHE, while the ensemble variants achieved the strongest overall results.

5.5 Class-wise Error Analysis

Macro F1 revealed important differences that would be hidden by accuracy alone. Table 8 compares class-wise F1 values for the strongest ViT configuration and the final V3 class-adjusted ensemble. The ensemble improved the balance across all five classes and achieved a notably higher F1 for class 1 than the ViT. However, class 1 remained difficult because the official test set contains only five samples from this grade.

Figure 5 shows the class-wise F1 scores and confusion matrices for the best preprocessing CNN and the final V3 ensemble. The ensemble improves the minority-class profile and

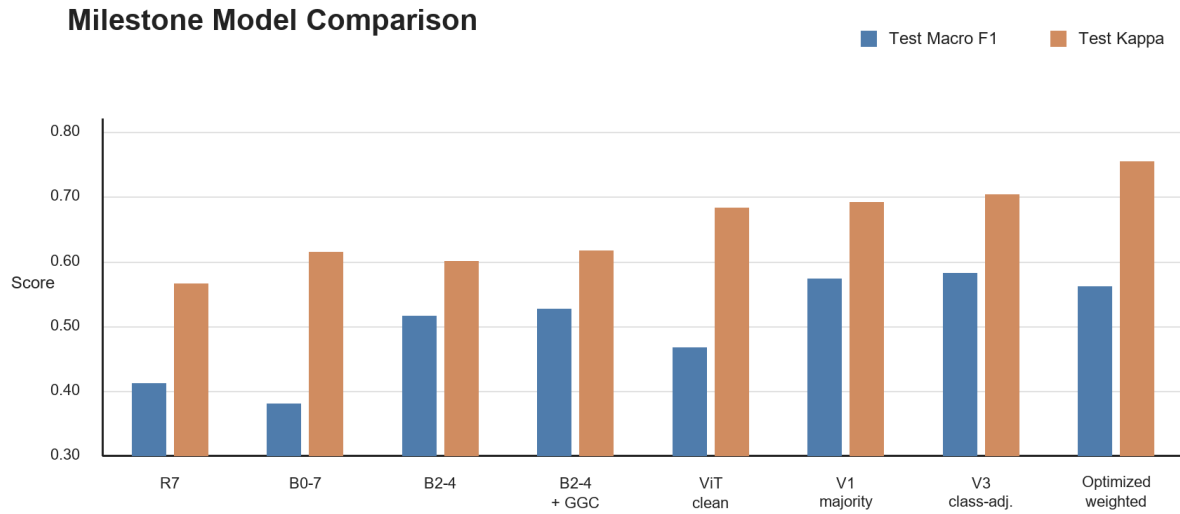


Figure 4. Milestone comparison of test Macro F1 and quadratic weighted kappa.

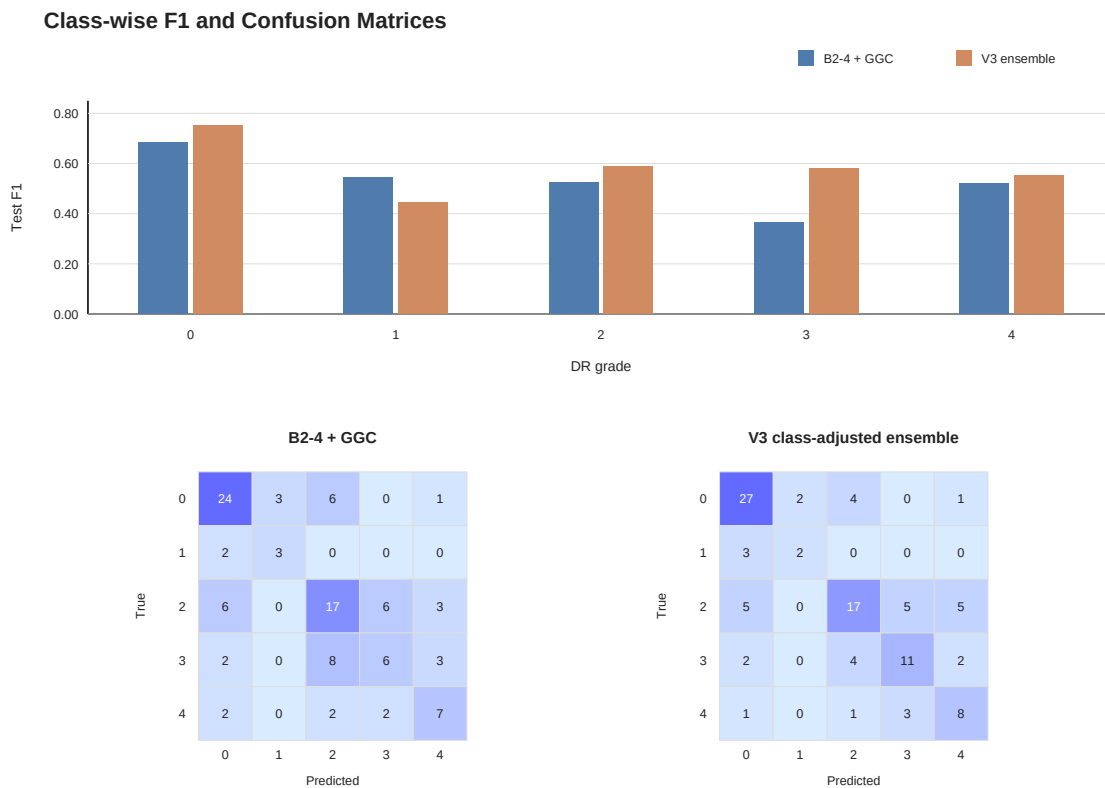


Figure 5. Class-wise F1 scores and confusion matrices for B2-4 with Gray-Gamma-CLAHE and the V3 class-adjusted ensemble.

produces a more balanced distribution of correct predictions across the five grades.

5.6 Qualitative Interpretation with Grad-CAM

Grad-CAM was used as a qualitative plausibility check for the EfficientNet-B2 model with Gray-Gamma-CLAHE. The goal was not to provide a clinical explanation, but to inspect whether the model focused on retinal image regions rather than only image borders or background artifacts. Figure 6 shows examples for a correctly classified grade-0 image, a correctly classified grade-4 image, a grade-2/3 confusion, and a class-1 example.

The visualizations indicate that the model often attends to regions inside the retinal field of view, including vascular and lesion-like areas. However, the heatmaps should be interpreted cautiously. Grad-CAM is only a qualitative plausibility check and does not prove that the model uses clinically meaningful features in every case.

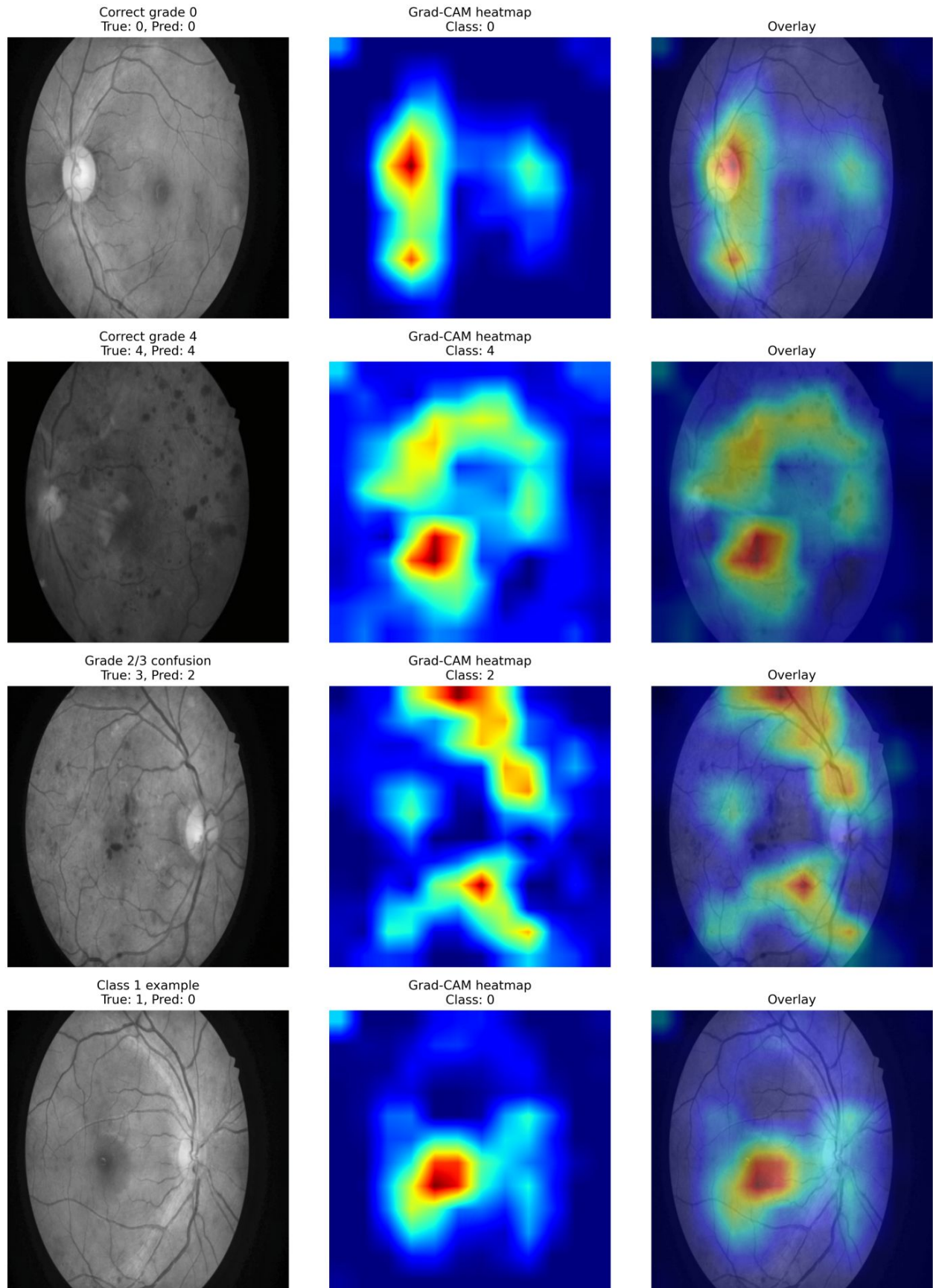


Figure 6. Grad-CAM examples for EfficientNet-B2 with Gray-Gamma-CLAHE. The visualizations provide a qualitative plausibility check of whether the model focuses on retinal image regions rather than image borders or background artifacts.

6. Discussion

The architecture comparison shows a clear advantage for EfficientNet-B2 among the individual models. The best original EfficientNet-B2 run, B2-4, reached a test Macro F1 of 0.5157, outperforming the best ResNet-18 run R7 with 0.4122 and the best EfficientNet-B0 run B0-7 with 0.3810. The additional capacity of EfficientNet-B2 and its compound scaling strategy may have helped the model capture more useful retinal image patterns than the smaller CNN baselines [7]. At the same time, the comparison should not be interpreted as a perfectly isolated architecture study, because the CNNs were trained at 384×384 pixels whereas the ViT-B/16 experiments used 224×224 pixels.

The Vision Transformer reached a lower test Macro F1 than the best EfficientNet-B2 model, but its best configuration still achieved a competitive quadratic weighted kappa of 0.6832. This suggests that the ViT sometimes preserved the ordinal structure of the grades even when class-wise F1 remained weaker. One likely reason is that ViTs rely heavily on pre-training and data scale to learn robust spatial relationships [8]. On a small medical dataset such as IDRiD, the stronger local inductive bias of CNNs may be advantageous, especially when lesions are small and local.

Across the CNN experiments, full fine-tuning was generally more effective than using a frozen backbone. The strongest ResNet-18 and EfficientNet-B0 runs were both fully fine-tuned configurations. This supports the idea that pretrained ImageNet features are useful as an initialization, but still need adaptation to the specific retinal grading task. Medical image analysis often benefits from fine-tuning rather than training from scratch or only replacing the final classifier, particularly when the target domain differs from natural images [10]. More generally, pretrained features are not equally transferable across all layers and tasks [11].

However, full fine-tuning also increases the risk of overfitting. This was visible in several validation-test gaps, especially for heavily tuned configurations. The ViT Optuna run is the clearest example: it reached a high validation score but produced a much weaker test Macro F1. This does not mean that Optuna or fine-tuning are unsuitable, but it shows that model selection on a validation set of only 83 images can become unstable.

The preprocessing experiments showed that more image processing was not automatically better. The original B2-4 model without manual preprocessing outperformed the early Crop+Graham variants. Later, Gray-Gamma-CLAHE improved B2-4 and produced the strongest individual Macro F1 of 0.5273. This indicates that targeted contrast enhancement can help, but only when it preserves useful lesion and color information. CLAHE-related techniques are designed to improve local contrast [16], and retinal preprocessing work also motivates illumination correction and enhancement for fundus

images [17, 18]. Nevertheless, aggressive preprocessing can remove or distort information that is relevant for classification.

Class imbalance remained a central challenge. Grade 1 is represented by very few samples, with only five examples in the official test set. Consequently, one or two changed predictions can noticeably affect the class-wise F1 and Macro F1. Weighted losses, focal loss, oversampling, and augmentation were therefore useful experimental factors, but none of them solved the imbalance problem completely. This matches the general difficulty of learning from imbalanced data [12]. Focal loss can emphasize difficult examples [20], but in this project the effects of imbalance strategies depended strongly on the model and preprocessing configuration. Some settings improved kappa while reducing Macro F1, suggesting a trade-off between ordinal agreement and balanced class-wise recognition.

The largest performance gain came from ensembles. The best individual model reached 0.5273 test Macro F1, while the V3 class-adjusted soft vote reached 0.5823. This improvement suggests that the individual models made complementary errors. Models with lower standalone Macro F1, such as EfficientNet-B0 or ResNet-18 variants, could still contribute useful predictions for samples that EfficientNet-B2 misclassified. This is consistent with the general motivation for deep ensembles, where diversity between models can improve robustness [22]. CNN ensembles have also been explored specifically for diabetic retinopathy grading [5].

Simple majority voting was already a strong baseline, which is important because it does not require optimized weights. Class-adjusted and optimized weighted variants improved some metrics, but they also increased the risk of fitting validation-specific noise. The logit-level fusion experiments did not outperform the strongest probability-based ensemble. Temperature scaling and logit normalization are useful calibration-related tools [21], but in this project they did not produce a better Macro F1 than the best soft-voting ensemble.

A recurring issue throughout the project was the gap between validation and test performance. The validation set contains only 83 images and only four grade-1 examples. This makes validation-based decisions noisy, especially for Optuna tuning and ensemble-weight optimization. A configuration can look strong on validation because it predicts a small number of rare-class samples correctly, but this may not generalize to the official test set.

For this reason, exploratory test-sorted findings were not treated as final model-selection results. The final ensemble comparison emphasizes validation-selected candidates. This distinction is important because repeated model selection can overfit the validation set and create an overly optimistic impression of performance [23]. Overall, the results should therefore be interpreted as evidence about trends and trade-offs rather than as a definitive ranking of architectures.

7. Limitations

This study has several limitations. First, the dataset is small and strongly imbalanced. In particular, grade 1 contains very few images in the validation and test splits. This makes class-wise metrics unstable and limits the reliability of conclusions about rare classes.

Second, the experiments use one primary train-validation split with seed 42. Repeated stratified cross-validation or multiple random seeds would provide a more robust estimate of model variance. This was not performed because of limited compute budget and Kaggle GPU constraints.

Third, the architecture comparison is not perfectly controlled. The CNNs were trained at 384×384 pixels, while the ViT-B/16 experiments used 224×224 pixels. The results therefore reflect the complete experimental pipelines rather than architecture alone.

Fourth, several parts of the project were exploratory. Optuna searches, preprocessing screening, and ensemble-weight optimization were useful for discovering promising configurations, but they also increase the risk of validation overfitting. This is especially relevant for the optimized ensemble variants.

Fifth, no external validation dataset was used. The official IDRiD test set provides a held-out evaluation, but the results may not transfer directly to fundus images from other cameras, populations, or annotation protocols.

Finally, Grad-CAM is only a qualitative plausibility check. It can indicate whether a CNN focuses on retinal regions rather than background artifacts, but it does not provide a clinical explanation of the prediction.

Future work should include repeated stratified splits, multiple seeds, and confidence intervals. Additional improvements could include ordinal loss functions, stronger retinal-domain pretrained models, external validation, and calibrated ensemble probabilities if calibration becomes a focus.

8. Conclusion

This report compared CNNs, a Vision Transformer, preprocessing strategies, imbalance handling, and ensemble methods for five-class diabetic retinopathy grading on IDRiD. EfficientNet-B2 was the strongest individual architecture. The best single model was B2-4 with Gray-Gamma-CLAHE, reaching a test Macro F1 of 0.5273. The best ViT-B/16 configuration, `clean_green_masked`, achieved a test Macro F1 of 0.4679 and a kappa of 0.6832.

The strongest overall Macro F1 was achieved by the V3 class-adjusted ensemble with 0.5823. The optimized weighted soft-vote ensemble achieved the highest observed kappa of 0.7546. These results indicate that model diversity and com-

plementary errors were more useful than simply increasing model complexity. At the same time, the experiments showed that small validation sets make hyperparameter tuning and ensemble selection unstable. Overall, the project demonstrates both the potential and the practical difficulty of deep learning for imbalanced medical image grading.

Appendix

This appendix contains the detailed result tables supporting the main report. Unless stated otherwise, all reported accuracy, Macro F1, and quadratic weighted kappa values refer to the official IDRiD test set. Macro F1 is abbreviated as F1 in the compact tables.

A. Complete Experiment Matrices

Table 9, Table 10, and Table 11 summarize the controlled CNN experiment matrices. The columns report whether the pretrained backbone was frozen, which preprocessing and imbalance strategy was used, the loss function, learning rate, batch size, best validation-selected epoch, and final test metrics.

Table 9. Complete ResNet-18 experiment matrix.

ID	Frozen	Preproc.	Sampling/Aug.	Loss	LR	BS	Epoch	Acc.	F1	Kappa
R1	Yes	None	None	CE	0.00010	4	19	0.4660	0.3444	0.5661
R2	Yes	None	None	Weighted CE	0.00010	4	20	0.4563	0.3471	0.5747
R3	Yes	Crop-Graham	None	CE	0.00010	4	8	0.4757	0.3659	0.6012
R4	Yes	Circular-Autocontrast	None	CE	0.00010	4	18	0.4660	0.3260	0.5073
R5	Yes	Crop-Graham	Weighted oversampling	CE	0.00010	4	14	0.4272	0.3725	0.5754
R6	Yes	Crop-Graham	Mild augmentation	CE	0.00010	4	8	0.4078	0.3140	0.4660
R7	No	Crop-Graham	None	CE	0.00003	4	42	0.5243	0.4122	0.5663

Table 10. Complete EfficientNet-B0 experiment matrix.

ID	Frozen	Preproc.	Sampling/Aug.	Loss	LR	BS	Epoch	Acc.	F1	Kappa
B0-1	Yes	None	None	CE	0.00100	8	12	0.4951	0.3565	0.4840
B0-2	Yes	None	None	Weighted CE	0.00100	8	29	0.4757	0.3115	0.5315
B0-3	Yes	Crop-Graham	None	CE	0.00100	8	7	0.3883	0.2899	0.5262
B0-4	Yes	Circular-Autocontrast	None	CE	0.00100	8	20	0.4660	0.3081	0.4935
B0-5	Yes	Crop-Graham	Weighted oversampling	CE	0.00100	8	19	0.4175	0.3113	0.3733
B0-6	Yes	Crop-Graham	Mild augmentation	CE	0.00100	8	7	0.4854	0.3646	0.6312
B0-7	No	Crop-Graham	None	CE	0.00010	8	3	0.5340	0.3810	0.6146

Table 11. Complete EfficientNet-B2 experiment matrix.

ID	Frozen	Preproc.	Sampling/Aug.	Loss	LR	BS	Epoch	Acc.	F1	Kappa
B2-1	No	Crop-Graham	None	CE	0.00005	4	26	0.5340	0.4257	0.5649
B2-2	Yes	Crop-Graham	Mild augmentation	CE	0.00050	4	4	0.4660	0.4058	0.5694
B2-3	No	Crop-Graham	Mild augmentation	CE	0.00005	4	14	0.5534	0.4139	0.5761
B2-4	No	None	None	CE	0.00005	4	4	0.5825	0.5157	0.6007
B2-5	No	None	Mild augmentation	CE	0.00005	4	21	0.5631	0.4976	0.6930
B2-6	No	None	None	Weighted CE	0.00005	4	21	0.5728	0.4877	0.6463

B. Preprocessing Screening

The additional preprocessing study used a two-stage procedure. First, several preprocessing methods were screened using the B2-4 configuration and validation Macro F1 only. Second, selected methods were confirmed with official test-set evaluation. The screening rows therefore contain validation metrics only by design.

Table 12. Validation-only preprocessing screening with EfficientNet-B2.

Run	Preprocessing	Best epoch	Val F1
screen-B2-4-none	None	4	0.5767
screen-B2-4-gamma	Gamma correction	5	0.6362
screen-B2-4-GGC	Gray-Gamma-CLAHE	13	0.6397
screen-B2-4-green-hist	Green histmatch median CLAHE unsharp crop	4	0.5655
screen-B2-4-GCBM	Green-CLAHE-BGSub-Mask	6	0.6152
screen-B2-4-clean-green	Clean green masked	10	0.5881
screen-B2-4-white-balance	White balance	7	0.5713
screen-B2-4-WB-CLAHE	White balance + CLAHE	3	0.5457

Table 13. Confirmed preprocessing runs with official test evaluation.

Run	Base	Preprocessing	Epoch	Val F1	Acc.	Test F1	Kappa
confirm-B2-4-GGC	B2-4	Gray-Gamma-CLAHE	7	0.5918	0.5534	0.5273	0.6169
confirm-B2-4-gamma	B2-4	Gamma correction	5	0.6362	0.5825	0.4452	0.6475
confirm-B2-5-GGC	B2-5	Gray-Gamma-CLAHE	9	0.5775	0.5340	0.4150	0.5940
confirm-B2-5-gamma	B2-5	Gamma correction	7	0.5691	0.5340	0.4439	0.6796
confirm-B2-6-GGC	B2-6	Gray-Gamma-CLAHE	12	0.5990	0.5437	0.4984	0.6967
confirm-B2-6-gamma	B2-6	Gamma correction	9	0.6137	0.4854	0.4429	0.6296
confirm-B2-4-GCBM	B2-4	Green-CLAHE-BGSub-Mask	6	0.6152	0.5631	0.5044	0.6681

C. Ensemble Search

The ensemble search combined strong CNN runs and preprocessing variants. Candidate models were selected using validation metrics, and the retained ensemble variants were then evaluated on the official test set. Table 14 lists the individual candidate models used in the final ensemble stage.

Table 14. Candidate models used for the final ensemble search.

ID	Model	Preprocessing	Variant	Val Acc.	Val F1	Val Kappa
B2-4	EfficientNet-B2	None	Original B2 run	0.7108	0.5767	0.8106
B2-5	EfficientNet-B2	None	Augmented B2 run	0.7229	0.6575	0.8226
B2-6	EfficientNet-B2	None	Weighted-loss B2 run	0.6867	0.6291	0.7940
R7	ResNet-18	Crop-Graham	Fully fine-tuned ResNet	0.7108	0.6145	0.8384
B0-7	EfficientNet-B0	Crop-Graham	Fully fine-tuned B0	0.6386	0.5752	0.7883
B0-6	EfficientNet-B0	Crop-Graham	Augmented B0 run	0.6747	0.5854	0.7535
B2-4-GGC	EfficientNet-B2	Gray-Gamma-CLAHE	Confirmed preprocessing run	0.6867	0.5918	0.8092
B2-6-GGC	EfficientNet-B2	Gray-Gamma-CLAHE	Confirmed preprocessing run	0.6506	0.5990	0.7694
B2-4-GCBM	EfficientNet-B2	Green-CLAHE-BGSub-Mask	Confirmed preprocessing run	0.7229	0.6152	0.8357

Table 15. Validation-selected ensemble finalists.

Method	Models	N	Val F1	Val Kappa	Test Acc.	Test F1	Kappa
Class-adjusted soft vote	B2-4, B2-5, B2-6, R7, B0-7, B0-6, B2-6-GGC	7	0.8172	0.9151	0.6311	0.5823	0.7036
Optimized weighted soft vote	B2-4, B2-6, R7, B0-7, B0-6, B2-6-GGC, B2-4-GCBM	7	0.7715	0.9214	0.6117	0.4764	0.6805
Majority vote	B2-4, R7, B0-7, B2-6-GGC, B2-4-GCBM	5	0.7526	0.8978	0.6117	0.5336	0.6943
Soft vote	B2-5, B0-7, B2-6-GGC	3	0.7204	0.8395	0.6117	0.4872	0.7089
Weighted soft vote	B2-5, B0-7, B2-6-GGC	3	0.7204	0.8395	0.6117	0.5443	0.7141

For the class-adjusted soft vote, the validation-optimized model weights were [0.1972, 0.0295, 0.0790, 0.2833, 0.1509, 0.0540, 0.2061] for the models in the order shown in Table 15. The class-adjustment vector was [0.8885, 1.6227, 0.9827, 1.4757, 1.2201].

Table 16. Selected probability and logit-fusion finalists.

Method	Fusion	Models	N	Val F1	Test Acc.	Test F1	Kappa
Optimized weighted soft vote	Probability	B2-5, R7, B0-7, B0-6, B2-4-GGC, B2-6-GGC	6	0.7666	0.6311	0.5616	0.7546
Optimized weighted logits	Raw logits	B2-5, B0-7, B2-6-GGC	3	0.7642	0.6214	0.5001	0.7088
Temperature optimized logits	Temperature logits	B2-4, B2-5, B2-6, R7, B0-7, B2-6-GGC	6	0.7640	0.6214	0.5400	0.6436
Z-score optimized logits	Z-score logits	B2-4, B2-5, R7, B0-7, B2-4-GGC, B2-6-GGC	6	0.7640	0.6311	0.5364	0.6817
Mean logits	Raw logits	B2-5, R7, B0-7, B2-6-GGC	4	0.7304	0.6214	0.5416	0.7149
Z-score mean logits	Z-score logits	B2-5, R7, B0-7, B2-6-GGC	4	0.7304	0.6214	0.5430	0.7070
Weighted soft vote	Probability	B2-5, B0-7, B2-6-GGC	3	0.7204	0.6117	0.5443	0.7141

The optimized weighted soft vote in Table 16 used the weights [0.2435, 0.0391, 0.2276, 0.0656, 0.0830, 0.3412], corresponding to the model order shown in the table.

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AI Use Statement

The author used GPT-5.5 as an AI-assisted tool during this project. Its use included support for code generation, debugging, report structuring, and language editing. The author reviewed and adapted all generated content and remain fully responsible for the experiments, results, interpretations, and final manuscript.